Resistance

Name _____

Measure the resistance in each of the following items using the multimeter. Use the 200 Ω scale

Make sure you make a good connection with the terminals

Resistance	value (Ω)
Unknown resistance 1	
Unknown resistance 2	
Unknown resistance 3	
Unknown resistance 4	
Unknown resistance 5	
Unknown resistance 6	
Unknown resistance 7	
12 V lamp (the large one)	
6 V lamp (the smaller one)	
Short electrical cord with 3 pin plug	
electric motor	
minimum resistance on rheostat	
maximum resistance on rheostat	
long conducting wire	
short conducting wire	
heater coil	

Questions

1 Which appliance among the ones you measured has the greatest resistance?

2 Which appliance among the ones you measured has the least resistance?

3 Why do the conducting wires have a low resistance compared to other items you have tested?

4 Why does the heating coil have a relatively large resistance?

5 Would you expect a clothes iron to have a high or low resistance?

6 Explain your answer to Q5 above.

7 In terms of the kinetic theory how you expect the particles in a high resistance wire to be behaving when an electric current is passed through the ghwire?